

Fourierreihe:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

$$\Rightarrow T = 2\pi$$

$$\Rightarrow \text{Koeffizienten: } \boxed{a_n = \frac{2}{T}} \quad T=2\pi \Rightarrow a_n = \frac{1}{\pi}$$

+ für beliebige Periodendauer T

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos\left(\frac{2\pi n x}{T}\right) + b_n \sin\left(\frac{2\pi n x}{T}\right)] \quad (3)$$

$$a_n = \frac{2}{T} \int_0^T f(x) dx \quad \Rightarrow \quad a_n = \boxed{\frac{2}{T}} \int_0^T f(x) dx$$

$\Rightarrow$ analog für b_n und b_n

komplexe Darstellung:

$$\cos(nx) = \frac{1}{2} (e^{jnx} + e^{-jnx}) \quad (1)$$

$$\sin(nx) = \frac{1}{2j} (e^{jnx} - e^{-jnx}) \quad (2)$$

$\Rightarrow$ einsetzen von (1) und (2) in (3)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \frac{1}{2} (e^{jnx} + e^{-jnx}) + b_n \frac{1}{2j} (e^{jnx} - e^{-jnx}) \right]$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} \frac{1}{2} (a_n - j b_n) e^{jnx} + \sum_{n=1}^{\infty} \frac{1}{2} (a_n + j b_n) e^{-jnx}$$

$$b_n \frac{1}{2j} = b_n \cdot \frac{1}{2j} = -j b_n \frac{1}{2}$$

$$n=0 \Rightarrow e^{j0x} = 1 \quad \Rightarrow \quad \frac{a_0}{2} = \frac{a_0}{2} \cdot 1 = \frac{a_0}{2} \cdot e^{j0x}$$

$$\Rightarrow f(x) = \sum_{n=-\infty}^{\infty} c_n e^{jnx} \quad \text{komplexe Darstellung}$$

$$\text{Bestimmung der Koeffizienten: } c_n = \frac{1}{T} \int_0^T f(x) e^{-jnx} dx$$

$$\text{Bestimmung: } c_n = \frac{2}{T}$$

$$a_n = 2c_n$$

$$\text{reelles } n \rightarrow c_n = \frac{1}{2} (a_n - j b_n) \quad a_n = c_n + \bar{c}_n$$

$$\text{virtuelles } n \rightarrow \bar{c}_n = \frac{1}{2} (a_n + j b_n) \quad b_n = j (c_n - \bar{c}_n)$$

Beispiel:

$$f(x) = \begin{cases} \frac{2\pi^2}{T} x & 0 \leq x < \frac{T}{2} \\ \frac{2\pi^2}{T} (x - T) & \frac{T}{2} \leq x < T \end{cases} \quad \begin{matrix} f = \text{Sawtooth} \\ T = \text{beliebig}$$

period. Sawtooth $\tau = T$ + alternierendes Signal

$$c_n = \frac{1}{T} \int_0^T f(x) e^{-jnx} dx$$

$$= \frac{1}{T} \left[\underbrace{\int_0^{\frac{T}{2}} \frac{2\pi^2}{T} x e^{-jnx} dx}_{I_1} + \underbrace{\int_{\frac{T}{2}}^T \frac{2\pi^2}{T} (x - T) e^{-jnx} dx}_{I_2} \right]$$

- partielle Integration für I_1 und I_2

$$I_1 = \int_0^{\frac{T}{2}} \frac{2\pi^2}{T} x e^{-jnx} dx = \frac{2\pi^2}{T} \int_0^{\frac{T}{2}} x e^{-jnx} dx = \frac{2\pi^2}{T} \left[-\frac{x e^{-jnx}}{jn} \right]_0^{\frac{T}{2}}$$

$$= -\frac{2\pi^2}{T} \left[\frac{x e^{-jnx}}{jn} \right]_0^{\frac{T}{2}} = -\frac{2\pi^2}{T} \left[\frac{\frac{T}{2} e^{-jn \frac{T}{2}}}{jn} - 0 \right]$$

$$I_2 = \int_{\frac{T}{2}}^T \frac{2\pi^2}{T} (x - T) e^{-jnx} dx = \frac{2\pi^2}{T} \int_{\frac{T}{2}}^T (x - T) e^{-jnx} dx$$

$$= \frac{2\pi^2}{T} \left[\frac{(x - T) e^{-jnx}}{jn} \right]_{\frac{T}{2}}^T \quad \begin{matrix} \text{Integration} \\ \text{in } (x) \end{matrix}$$

$$= \frac{2\pi^2}{T} \left(\left[\frac{(T - T) e^{-jnT}}{jn} \right] - \left[\frac{(\frac{T}{2} - T) e^{-jn \frac{T}{2}}}{jn} \right] \right) \quad \begin{matrix} \text{Integration} \\ \text{in } (x) \end{matrix}$$

$$= \frac{2\pi^2}{T} \left(\left[\frac{0}{jn} \right] - \left[\frac{-\frac{T}{2} e^{-jn \frac{T}{2}}}{jn} \right] \right) = \frac{2\pi^2}{T} \left[\frac{\frac{T}{2} e^{-jn \frac{T}{2}}}{jn} \right]$$

$$e^{-jn \frac{T}{2}} = e^{-j \frac{2\pi n}{T} \frac{T}{2}} = \cos\left(\frac{2\pi n}{2}\right) - j \sin\left(\frac{2\pi n}{2}\right) = 1$$

$$e^{-jnT} = e^{-j \frac{2\pi n}{T} T} = \cos(2\pi n) - j \sin(2\pi n) = 1$$

mit diesen Formeln ergibt sich:

$$I_1 = \frac{2\pi^2}{T} \left[\frac{\left(\frac{T}{2} \right) (1 - 1)}{jn} \right] = \frac{2\pi^2}{T} \left[\frac{0}{jn} \right]$$

$$= \frac{2\pi^2}{T} \left[\frac{0}{jn} \right] = 0$$

$$I_2 = -\frac{2\pi^2}{T} \left[\frac{(1 - 1) e^{-jnT}}{jn} \right] = \frac{2\pi^2}{T} \left[\frac{(1 - 1) e^{-jnT}}{jn} \right]$$

$$= \frac{2\pi^2}{T} \left[\frac{(1 - 1) e^{-jnT}}{jn} \right] = \frac{2\pi^2}{T} \left[\frac{0}{jn} \right] = 0$$

da wir die beiden Integrale mit $\frac{1}{T}$ multiplizieren und addieren:

$$c_n = \frac{1}{T} (I_1 + I_2) = \frac{1}{T} \left[\frac{2\pi^2}{T} \left(\frac{(1 - 1) e^{-jnT}}{jn} \right) + \frac{2\pi^2}{T} \left(\frac{(1 - 1) e^{-jnT}}{jn} \right) \right]$$

$$= \frac{1}{T} \left[\frac{2\pi^2}{T} \left(\frac{(1 - 1) e^{-jnT}}{jn} \right) + \frac{2\pi^2}{T} \left(\frac{(1 - 1) e^{-jnT}}{jn} \right) \right]$$

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$$= \frac{1}{T} \left[\frac{2\pi^2}{T} \left(\frac{(1 - 1) e^{-jnT}}{jn} \right) + \frac{2\pi^2}{T} \left(\frac{(1 - 1) e^{-jnT}}{jn} \right) \right]$$

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